



COURSE BROCHURE

Prepared By
The Team @Trainingcred



TRANSACTION-SQL AND PYTHON PROGRAMMING TRAINING COURSE

COURSE OVERVIEW

Modern data professionals must demonstrate measurable impact through efficient query design, automated data processing, reliable performance optimization, scalable integration solutions, and clear reporting of analytical findings. Whether you're managing enterprise databases, supporting business intelligence initiatives, analyzing customer behavior patterns, optimizing supply chain operations, or building data-driven applications, stakeholders expect you to show current system performance, identify bottlenecks and inefficiencies, set realistic improvement targets, implement high-impact optimizations, and track measurable results.

This course provides a systematic approach to mastering both Transaction-SQL and Python programming as complementary tools for comprehensive data management. You'll gain expertise in advanced T-SQL query optimization, stored procedure development, database automation, Python data manipulation libraries, statistical analysis capabilities, API integration, database connectivity, and enterprise deployment strategies. The methodology is hands-on and outcome-driven, designed for professionals who must deliver reliable solutions under real-world constraints.

We acknowledge the reality of your working environment: legacy system limitations, competing performance requirements, tight deployment schedules, budget constraints for new tools, and the need to maintain existing operations while implementing improvements. This course is designed for professionals who must deliver measurable results within these constraints, not in idealized laboratory conditions.

TARGET AUDIENCE

This course is designed for professionals who are directly responsible for, or accountable for, database management, data analysis, and automated data processing across their organizations.

This course is designed for:

- Database Administrators responsible for query optimization, performance tuning, and automated maintenance procedures
- Data Analysts who need to extract, transform, and analyze complex datasets using both SQL and Python
- Business Intelligence Developers tasked with creating reports, dashboards, and automated data workflows
- Software Developers who integrate database operations with application logic and analytics capabilities
- Data Scientists requiring strong foundational skills in data extraction, manipulation, and statistical analysis
- IT Managers overseeing database operations, data quality initiatives, and analytics infrastructure
- Systems Analysts responsible for designing and optimizing data processing workflows and reporting systems
- Application Developers who build data-driven applications requiring efficient database integration
- Quality Assurance Engineers who validate data integrity, test database procedures, and automate testing workflows
- Anyone accountable for improving database performance, automating data processes, or delivering analytical insights through technical implementation

LEARNING OBJECTIVES

This course equips you to design, implement, and optimize database solutions that integrate advanced T-SQL querying with Python automation, deliver measurable performance improvements, and support enterprise-scale analytical requirements.

By the end of this course, you'll be able to:

- Understand advanced T-SQL concepts, query optimization principles, and Python programming fundamentals for comprehensive data management
- Measure database performance metrics, identify bottlenecks, and quantify the impact of optimization strategies across complex query workloads
- Design efficient stored procedures, functions, and triggers that automate routine database operations while maintaining data integrity and performance standards
- Apply Python data manipulation libraries to extract, transform, and analyze data from multiple sources with robust error handling and logging
- Develop integrated solutions that combine T-SQL database operations with Python analytics, machine learning, and automated reporting capabilities
- Assess data quality issues, implement validation procedures, and create monitoring systems that ensure ongoing data reliability and accuracy
- Set performance targets, establish monitoring dashboards, and implement optimization strategies that deliver measurable improvements in query execution and system throughput
- Communicate technical solutions effectively to stakeholders, document procedures for maintenance teams, and train colleagues on implemented systems and workflows

REQUIREMENTS & PREREQUISITES

Participants should have basic SQL query experience and fundamental programming concepts. Familiarity with relational database concepts is essential. No prior Python experience required, but logical thinking and problem-solving skills are important. Access to SQL Server (or equivalent) and Python development environment recommended for hands-on exercises.

PROFESSIONAL AND ORGANIZATIONAL IMPACT

When you lead database operations and analytics with credible technical skills and proven optimization strategies, you become a trusted driver of operational efficiency and data-driven decision making.

As a participant, you will benefit by:

- Build expertise in both T-SQL advanced querying and Python programming, positioning yourself as a versatile data professional capable of end-to-end solution development
- Gain confidence in performance tuning, query optimization, and system troubleshooting that enables you to resolve complex technical challenges independently
- Strengthen your ability to balance database performance, system reliability, and analytical requirements while meeting tight project deadlines and budget constraints
- Enhance credibility with development teams, business analysts, and IT leadership through demonstrated ability to deliver measurable performance improvements
- Develop skills in automation and workflow optimization that reduce manual effort and increase your capacity for strategic, high-value projects
- Position yourself as a data solutions architect as organizations increasingly seek professionals who can bridge database management and modern analytics
- Expand career opportunities in data engineering, business intelligence, analytics, and database administration as demand grows for integrated technical skills

Organizations that embed advanced database programming and Python analytics capabilities into their data operations reduce costs, mitigate risks, and build lasting competitive advantage through faster, more reliable insights.

Your organization will benefit from:

- Direct cost savings through optimized query performance, reduced server resource consumption, and automated processes that eliminate manual data handling overhead
- Improved data quality and regulatory compliance through systematic validation procedures, error handling, and audit trail capabilities built into automated workflows
- Enhanced decision-making speed and accuracy through integrated analytics solutions that combine database efficiency with advanced Python-based statistical analysis
- Reduced system downtime and performance issues through proactive monitoring, automated maintenance procedures, and optimized database operations
- Increased team productivity through standardized procedures, reusable code libraries, and automated reporting systems that free analysts for higher-value work
- Stronger competitive positioning through faster time-to-insight, more sophisticated analytical capabilities, and reliable data infrastructure that supports business growth
- Better return on technology investments through maximized utilization of existing database systems and strategic integration with modern Python analytics tools

TRAINING METHODOLOGY

This is a practical, outcome-driven course designed to turn database programming aspirations into measurable technical proficiency and reliable system improvements.

Methodology includes:

- Guided query optimization exercises using real-world datasets with performance benchmarking and measurable improvement tracking
- Hands-on Python programming labs focused on database connectivity, data manipulation, and automated workflow development with error handling
- Database performance assessment using industry-standard monitoring tools and optimization techniques applicable across different database environments
- Integration project combining T-SQL stored procedures with Python analytics scripts to create end-to-end automated reporting solutions
- Industry-specific case studies from financial services, healthcare, retail, manufacturing, and logistics demonstrating practical implementation approaches
- Collaborative troubleshooting sessions where participants diagnose performance issues, design solutions, and validate improvements under realistic constraints
- Professional development exercises that challenge current practices and guide participants toward industry best practices for maintainable, scalable solutions

COURSE OUTLINE

MODULE 1: DATABASE PROGRAMMING FUNDAMENTALS AND ENVIRONMENT SETUP

- T-SQL architecture, query execution principles, and optimization fundamentals in enterprise database environments
- Python programming basics, data types, control structures, and best practices for database integration projects
- Development environment setup including SQL Server Management Studio, Python IDEs, and connection libraries configuration
- Version control, documentation standards, and project structure for maintainable database programming solutions
- Exercise: Configure integrated development environment and establish connection between Python and SQL Server database

MODULE 2: ADVANCED T-SQL QUERY DESIGN AND OPTIMIZATION

- Complex JOIN operations, subqueries, CTEs, and window functions for sophisticated data analysis and reporting
- Query execution plans, index utilization strategies, and performance tuning methodologies for large-scale databases
- Advanced data manipulation using MERGE statements, table expressions, and bulk operations for efficient data processing
- Error handling, transaction management, and concurrency control for robust database operations
- Exercise: Optimize poorly performing queries and document measurable performance improvements with execution plan analysis

MODULE 3: STORED PROCEDURES, FUNCTIONS, AND DATABASE AUTOMATION

- Stored procedure design patterns, parameter handling, and return value strategies for flexible, reusable database operations
- User-defined functions, table-valued functions, and scalar functions for modular data processing and business logic implementation
- Triggers for data validation, audit trails, and automated business rule enforcement across database operations
- Dynamic SQL generation, SQL injection prevention, and secure coding practices for robust database applications
- Exercise: Develop automated data validation stored procedure with comprehensive error handling and logging capabilities

MODULE 4: PYTHON DATA MANIPULATION AND DATABASE CONNECTIVITY

- Python database connectivity using pyodbc, SQLAlchemy, and pandas for efficient data extraction and manipulation
- Pandas DataFrame operations, data cleaning techniques, and transformation workflows for analytical data preparation
- NumPy arrays, mathematical operations, and statistical calculations for quantitative analysis and data validation
- File I/O operations, CSV/Excel processing, and data format conversions for integrated data workflows
- Exercise: Create Python script that extracts data from SQL Server, performs transformations, and loads results back to database

MODULE 5: STATISTICAL ANALYSIS AND DATA VISUALIZATION WITH PYTHON

- Statistical analysis using SciPy, descriptive statistics, hypothesis testing, and correlation analysis for data insights
- Data visualization with Matplotlib and Seaborn for creating professional charts, graphs, and analytical dashboards
- Machine learning fundamentals using Scikit-learn for predictive modeling and pattern recognition in business data
- Time series analysis techniques for forecasting, trend analysis, and seasonal pattern identification
- Exercise: Develop comprehensive analytical report combining T-SQL data extraction with Python statistical analysis and visualization

MODULE 6: AUTOMATED REPORTING AND WORKFLOW INTEGRATION

- Task scheduling using Windows Task Scheduler, cron jobs, and Python scheduling libraries for automated workflow execution
- Email automation, report distribution, and notification systems for stakeholder communication and alert management
- Template-based reporting using Jinja2, automated document generation, and professional output formatting
- Configuration management, environment variables, and deployment strategies for production automation systems
- Exercise: Build automated monthly reporting system that extracts data via T-SQL, processes with Python, and distributes formatted reports

MODULE 7: API INTEGRATION AND EXTERNAL DATA SOURCES

- RESTful API consumption using Python requests library, authentication handling, and rate limiting strategies
- JSON and XML data processing, parsing techniques, and integration with relational database structures
- Web scraping fundamentals using BeautifulSoup and Scrapy for automated data collection from web sources
- Data source validation, quality assessment, and integration testing for reliable external data incorporation
- Exercise: Create integration solution that pulls data from external API, validates quality, and stores in SQL Server database

MODULE 8: PERFORMANCE MONITORING AND SYSTEM OPTIMIZATION

- Database performance monitoring using DMVs, Query Store, and Extended Events for proactive system management
- Python application profiling, memory management, and optimization techniques for efficient script execution
- Logging frameworks, error tracking, and debugging strategies for maintainable production systems
- Capacity planning, scalability assessment, and performance baseline establishment for growing data volumes
- Exercise: Design comprehensive monitoring dashboard that tracks both database performance and Python workflow execution metrics

MODULE 9: SECURITY, COMPLIANCE, AND BEST PRACTICES

- Database security principles, role-based access control, and SQL injection prevention in stored procedures and dynamic queries
- Python security considerations, credential management, and secure coding practices for database applications
- Data privacy regulations, audit trail implementation, and compliance documentation for regulated industries
- Code review processes, testing strategies, and quality assurance procedures for reliable database programming solutions
- Exercise: Conduct security assessment of existing database procedures and Python scripts, implementing recommended security enhancements

MODULE 10: ENTERPRISE DEPLOYMENT AND STRATEGIC PLANNING

- Deployment planning, environment management, and release procedures for database and Python application changes
- Change management strategies, user training programs, and adoption planning for new automated systems
- Performance KPI definition, success metrics establishment, and ROI calculation for database programming investments
- Strategic roadmap development, technology evaluation, and future capability planning for evolving data requirements
- Exercise: Create comprehensive deployment plan and success metrics framework for organization-wide database programming initiative

CERTIFICATION

Participants who complete the Transact-SQL and Python Programming Training Program earn a Trainingcred Certificate of Achievement, demonstrating their professional competence and alignment with global standards in learning and development. Each certification reflects practical expertise, strategic insight, and readiness to excel in today's competitive, fast-evolving workplace.

Trainingcred Institute is a global provider of corporate training and EdTech solutions, offering a comprehensive range of programs across all major business functions, including Finance, Human Resources, Procurement and Supply Chain, Technology and Innovation, Quality, Operations, Engineering, Marketing, Sales, Customer Experience, Leadership, Strategy, Risk and Compliance, Data Analytics, and Sustainability.

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